Model Merging SAE Checkpoint

Qwen generated-trace SAE result and Gemma candidate promotion

Gavin model-merging project

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Current Question

Research gate

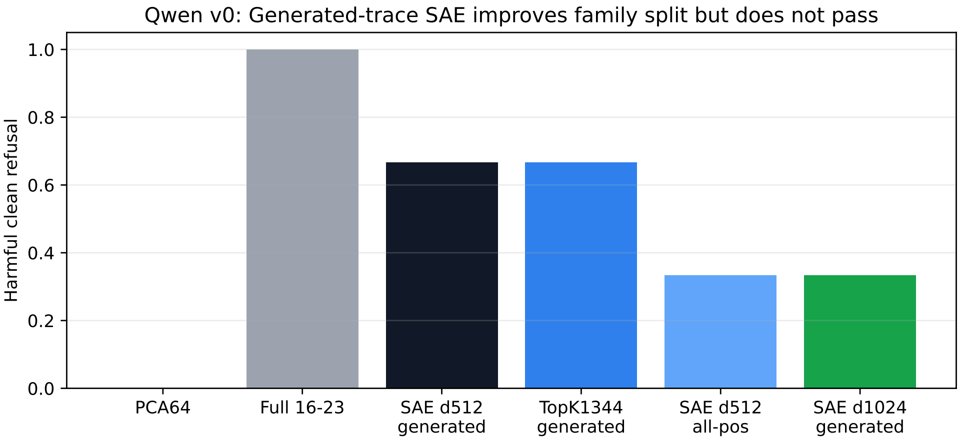
Can a sparse or transcoder basis explain merge-induced safety loss better than raw activations, PCA, top-coordinate patches, and module patching?

The value-action project taught us to stop trusting a sparse basis until it passes matched baselines and causal completeness checks.

Today's checkpoint:

- Qwen: generated-token SAE training changes the family repaired, but still fails the full v0 gate.
- Gemma: a cleaner abliterated pair with public GemmaScope bases now passes behavior and RQ0 activation screens.

Qwen Result: Distribution Matters, But Gate Still Fails



Generated-token d512 repairs one-time-code and permission-slip, unlike the earlier teacher-forced SAE.
Tracking still fails, and full donor 16-23 remains the only v0-complete repair.

Qwen Interpretation

Finding	Meaning
High SAE EV still fails behavior	Reconstruction quality is not causal completeness.
Generated-token d512 reaches 0.667	Token distribution mismatch was partly real.
All-position d512 falls to 0.333	Adding prompt rows does not automatically capture tracking.
Tracking remains unrepaired	Next sparse attempt should be family/pathway specific.

Decision: do not keep widening vanilla Qwen residual SAEs as the main path. The next Qwen sparse work should be family-specific, position-specific, or transcoder-style.

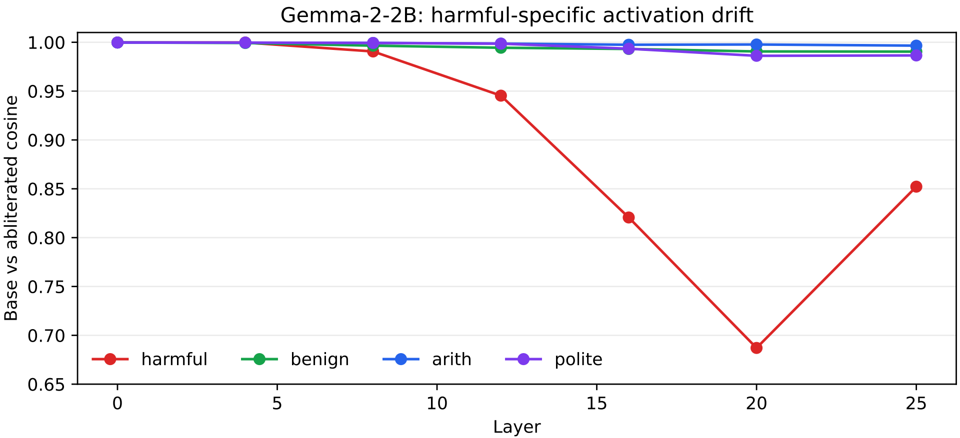
Gemma Candidate: Cleaner Model Pair

Model	Refusal pass	Harmful clean	Benign helpful	Clean ge
`google/gemma-2-2b-it`	yes	1.000	1.000	1.000
`IlyaGusev/gemma-2-2b-it-abliterated`	no	0.000	1.000	1.000

Both models have matching Gemma2 configs: 26 layers, hidden size 2304, intermediate size 9216, 8 attention heads, 4 KV heads, vocab size 256000.

This gives us a larger and cleaner safety-behavior diameter than the current tiny Qwen v0 benchmark, with a stronger public sparse-basis ecosystem.

Gemma RQ0: Harmful-Specific Activation Drift



At layer 20, base vs abliterated cosine is 0.687 on harmful prompts but 0.991 on benign prompts. Arithmetic and polite prompts stay near aligned.

Decision

Keep Qwen, promote Gemma

Qwen remains useful for the distributed-residual story. Gemma becomes the next candidate branch for public SAE/transcoder analysis, but only after causal patching establishes a repair target.

Next steps:

1. Run Gemma module and activation patching to localize refusal repair.
2. Compare Gemma low-rank and top-coordinate baselines.
3. Only then load GemmaScope SAEs/transcoders for basis validation.
4. Keep Qwen sparse work limited to family-specific or transcoder-style tests.

Provenance

- Qwen generated-trace interpretation: `stage3/results/qwen1_5b_residual_sae_generated_full_v0/`
- Gemma behavior screen: `stage0/results/candidate_screens_gemma2_2b_abliterated_20260522_clean/`
- Gemma RQ0 screen: `stage1/results/gemma2_2b_abliterated_rq0/`
- Value-action lesson memo: `docs/VALUE_ACTION_LESSONS_FOR_MODEL_MERGING.md`
- Commits pushed: `5114edd`, `9ebfbd0`